

Question 1 – Amounts (Bar Chart)

Scenario:A supermarket chain wants to compare the monthly sales revenue generated by different product categories during June 2026. The management wants to identify the best and worst performing categories using an appropriate visualization.

Code:

```
import matplotlib.pyplot as plt

categories = [
    "Groceries", "Dairy", "Fruits", "Vegetables", "Bakery",
    "Beverages", "Snacks", "Personal Care",
    "Household Items", "Frozen Foods"
]

sales = [120, 95, 80, 75, 68, 90, 110, 55, 70, 60]

plt.figure(figsize=(10,6))
plt.bar(categories, sales)

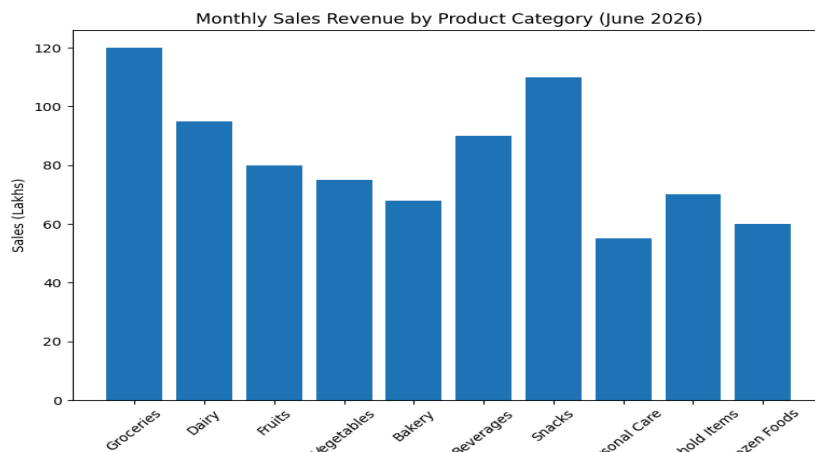
plt.title("Monthly Sales Revenue by Product Category (June 2026)")
plt.xlabel("Product Category")
plt.ylabel("Sales (Lakhs)")
plt.xticks(rotation=45)

plt.show()

# Highest and Lowest
highest = max(sales)
lowest = min(sales)

print("Highest Selling Category:", categories[sales.index(highest)], "-", highest, "Lakhs")
print("Lowest Selling Category:", categories[sales.index(lowest)], "-", lowest, "Lakhs")
```

Output :



Question 2 – Distribution (Histogram)

Code:

```
import matplotlib.pyplot as plt
```

```
salary = [3.2, 4.5, 5.0, 6.8, 7.2, 8.5, 4.8, 5.5, 6.0, 7.8, 9.2, 10.5]
```

```
plt.figure(figsize=(8,5))
```

```
plt.hist(salary, bins=5)
```

```
plt.title("Distribution of Salary Packages")
```

```
plt.xlabel("Salary Package (LPA)")
```

```
plt.ylabel("Number of Students")
```

```
plt.show()
```

Output:



Question 3 – Proportions (Pie Chart)

Code:

```
import matplotlib.pyplot as plt

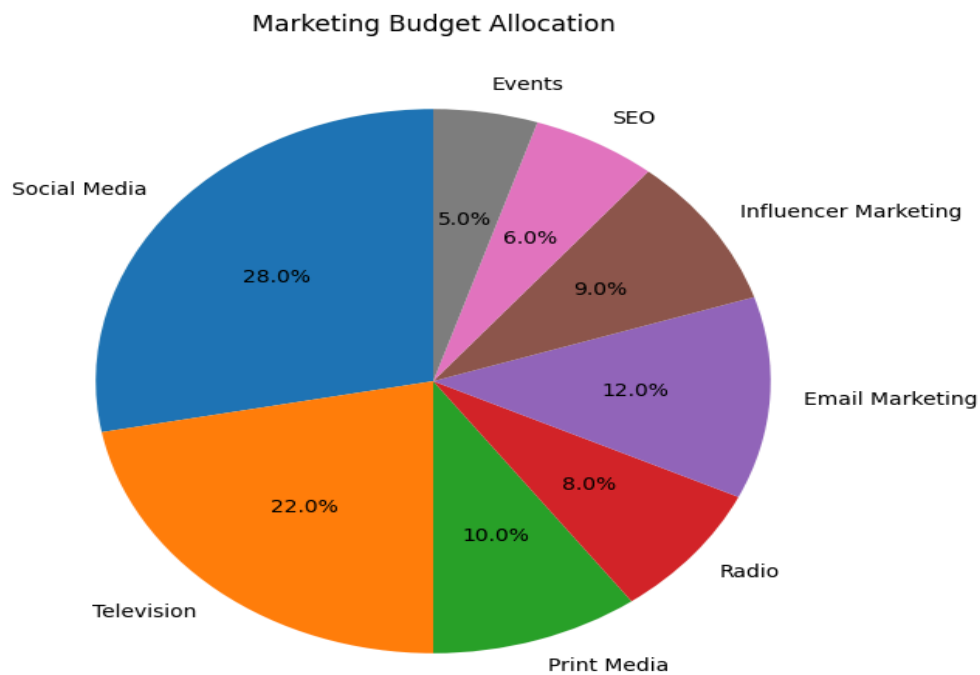
channels = [
    "Social Media",
    "Television",
    "Print Media",
    "Radio",
    "Email Marketing",
    "Influencer Marketing",
    "SEO",
    "Events"
]

budget = [28, 22, 10, 8, 12, 9, 6, 5]

plt.figure(figsize=(8,8))
plt.pie(
    budget,
    labels=channels,
    autopct="%1.1f%%",
    startangle=90
)

plt.title("Marketing Budget Allocation")
plt.show()
```

Output:



Question 4 – X–Y Relationship (Scatter Plot)

Code:

```
import matplotlib.pyplot as plt

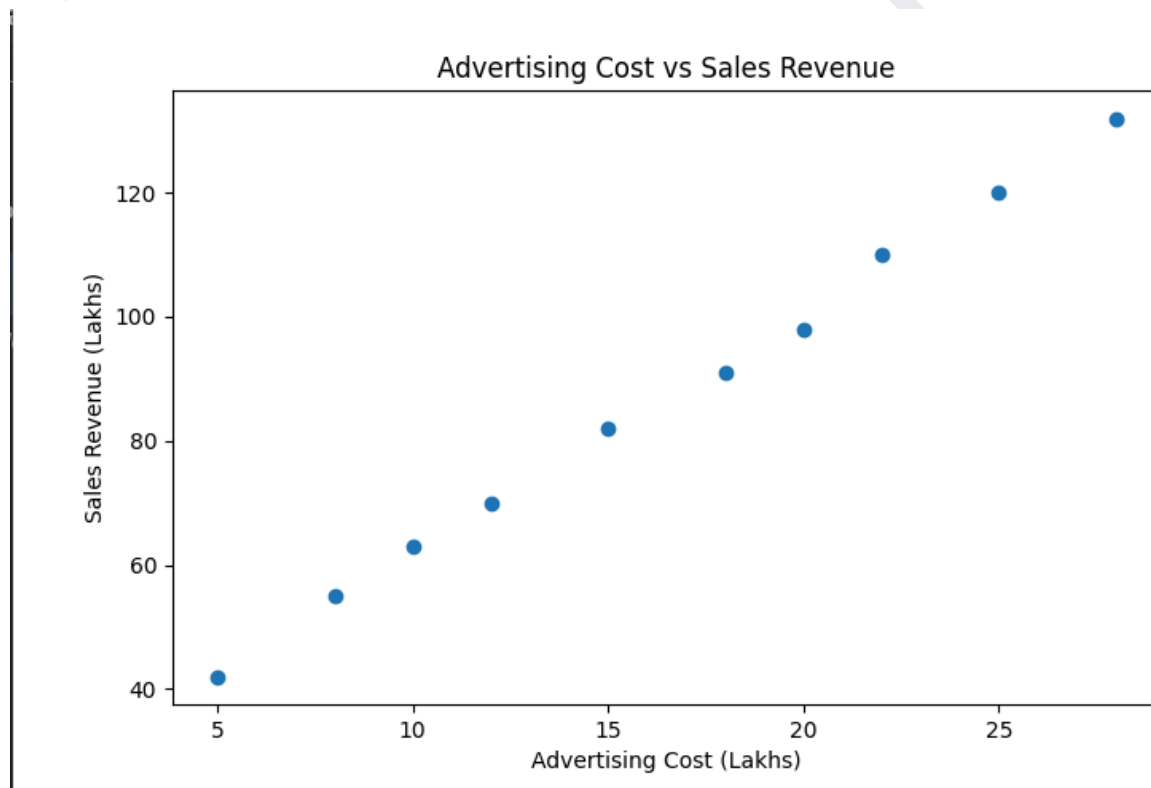
advertising = [5, 8, 10, 12, 15, 18, 20, 22, 25, 28]
sales = [42, 55, 63, 70, 82, 91, 98, 110, 120, 132]

plt.figure(figsize=(8,5))
plt.scatter(advertising, sales)

plt.title("Advertising Cost vs Sales Revenue")
plt.xlabel("Advertising Cost (Lakhs)")
plt.ylabel("Sales Revenue (Lakhs)")

plt.show()
```

Output:



Question 5 – Amounts Over Time (Line Chart)

Code:

```
import matplotlib.pyplot as plt

months = [
    "Jan", "Feb", "Mar", "Apr", "May", "Jun",
    "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"
]

energy = [420, 450, 490, 530, 560, 600, 580, 590, 610, 630, 590, 540]

plt.figure(figsize=(10,5))
plt.plot(months, energy, marker='o')

plt.title("Monthly Electricity Generation (2025)")
plt.xlabel("Months")
plt.ylabel("Energy Generated (MWh)")
plt.grid(True)

plt.show()

print("Highest Energy Production:", max(energy), "MWh in", months[energy.index(max(energy))])
print("Lowest Energy Production:", min(energy), "MWh in", months[energy.index(min(energy))])
```

Output:

